As a Senior Database Architect, I have designed the schema for the Banking Management System, prioritizing transactional integrity, auditability, and scalability.

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1. Database Overview

The database uses a relational model (RDBMS) to ensure ACID compliance for financial operations. It is structured to separate identity management, core banking services, and audit trails.

2. ER Diagram Description

- **User/Auth Core:** Centralizes users, roles, and MFA tokens.
- **Banking Core:** Manages accounts and transactions.
- **Loan/Workflow:** Manages application states and staff assignments.
- **Analytics/Fraud:** Stores AI insights and fraud logs.
- **Audit Trail:** Immutable logs linked to system actors.

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3. Tables & Schema

Core User & Identity

- **Users:** `user_id` (PK), `email`, `password_hash`, `role_id` (Admin/Employee/Customer), `mfa_secret`, `status`.
- **Roles:** `role_id` (PK), `role_name`.

Banking

- **Accounts:** `account_id` (PK), `user_id` (FK), `balance`, `currency`, `account_type`.
- **Transactions:** `transaction_id` (PK), `from_account` (FK), `to_account` (FK), `amount`, `timestamp`, `status`, `is_flagged`.

Loans

- **LoanApplications:** `loan_id` (PK), `user_id` (FK), `amount`, `monthly_income`, `status`, `processed_by` (FK - Employee ID).

Fraud & Insights

- **FraudLogs:** `log_id` (PK), `transaction_id` (FK), `reason`, `severity`, `status`.
- **AIInsights:** `insight_id` (PK), `user_id` (FK), `recommendation_text`, `generated_at`.

Audit

- **AuditLogs:** `audit_id` (PK), `user_id` (FK), `action`, `table_affected`, `old_value`, `new_value`, `timestamp`.

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4. SQL Schema (PostgreSQL Syntax)

```
CREATE TABLE users (
    user_id SERIAL PRIMARY KEY,
    email VARCHAR(255) UNIQUE NOT NULL,
    password_hash TEXT NOT NULL,
    role VARCHAR(50) NOT NULL,
    mfa_enabled BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE accounts (
    account_id SERIAL PRIMARY KEY,
    user_id INT REFERENCES users(user_id),
    balance DECIMAL(15, 2) DEFAULT 0.00,
    currency VARCHAR(3) DEFAULT 'USD',
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE transactions (
    transaction_id UUID PRIMARY KEY,
    from_account INT REFERENCES accounts(account_id),
    to_account INT REFERENCES accounts(account_id),
    amount DECIMAL(15, 2) NOT NULL,
    timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    status VARCHAR(20),
    is_flagged BOOLEAN DEFAULT FALSE
);

CREATE TABLE loan_applications (
    loan_id SERIAL PRIMARY KEY,
    user_id INT REFERENCES users(user_id),
    requested_amount DECIMAL(15, 2),
    status VARCHAR(20) DEFAULT 'SUBMITTED', -- SUBMITTED, PENDING, APPROVED, REJECTED
    processed_by INT REFERENCES users(user_id),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE audit_logs (
    audit_id SERIAL PRIMARY KEY,
    user_id INT REFERENCES users(user_id),
    action TEXT,
    timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

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5. Constraints & Indexes

- **Constraints:**

`*CHK_LoanLimit*`*: Trigger-based check ensuring* `*requested_amount* <= *(monthly_income* 5)`.

* `FK_Transactions`: Prevents deletion of accounts with existing transaction history.

- **Indexes:**

* `idx_user_email`: Optimized login performance.

* `idx_transaction_date`: Optimized reporting for audit and fraud detection.

* `idx_loan_status`: Optimized for Employee application queues.

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6. Normalization Notes

- **3NF Compliance:** All tables are in Third Normal Form.
- **Data Redundancy:** Minimal; customer profiles are normalized out of the `Accounts` table.
- **Auditability:** `AuditLogs` are designed to be append-only, ensuring the "Immutable logs" requirement is met.
- **Financial Precision:** `DECIMAL(15,2)` is used instead of `FLOAT` to prevent floating-point rounding errors common in financial calculations.

7. Recommendations

1. **Partitioning:** Partition the `Transactions` table by `timestamp` (Monthly) to ensure the 2-second performance requirement as data grows.
2. **Encryption:** Apply Column-Level Encryption on `balance` and `email` fields to satisfy security requirements.
3. **Soft Deletes:** Use `is_active` flags instead of `DELETE` commands to maintain data integrity for regulatory audits.